# Assignment 6: Replacement Gate Process and Metrology

## Starting resources

1. Files:
    a. ReplacementGate.zam

## Questions

At the end of this assignment, we expect a report. Add images or graphs to support your conclusions and answers.

1. Please open V2_ReplacementGate.zam. Look at the cross-section. Look at the process steps. Notice that it consists of a gate last process example as the gate metal stack is deposited AFTER the source and drain formation. Please notice that the device stops working if the metal contamination exceeds $10^{12}$ cm-3.
    a. The polysilicon dummy gate is refractory, so does not diffuse. Al diffusion only occurs during subsequent high-temperature steps. We have added a dummy anneal at the end to represent the combined effect of all anneal steps. Plot the metal diffusion in the structures after the dummy anneal. What is the Al concentration in the active region? Use the 'Aluminium Profile Export' analysis result to compute the average concentration in the active region. Is the device working?
2. Metrology of Replacement Gate process (RMG_metrology.vanalysis)
    a. What are the process parameters that affect the Replacement Gate process most?
    b. Use virtual metrology to plot Gate CD versus Gate etch parameters.
    c. Give two examples of how to decrease the Gate CD.
    d. Go to the gate lithography step. Introduce 5nm lithography bias. Run the process. Is the device working?
    e. Undo the process change of 2.d. For what etch and/or deposition parameters Replacement Gate process fails, notice void formation.
    f. Use the 'Void metrology' analysis result to measure the length, depth and area of the void.
    g. Go to the lithography step when opening the Source and Drain region. Change the lithography mask with the 'Source_Drain2' mask. Introduce a 10nm overlay (in X direction) in the lithography step of the gate. Run the process. Is the device still working?